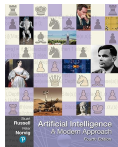


Lesson 08.4:

Instructor: Dr. GP Saggese, gsaggese@umd.edu

References:

- AIMA (Artificial Intelligence: a Modern Approach)
- Pearl et al., The Book of Why, 2017
- Facure, Causal Inference in Python, 2023



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WINNER OF THE TURING AWARD
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Sales example

- There are stores that put product on sales for Christmas

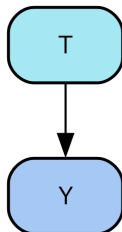
	store	weeks_to_xmas	avg_week_sales	is_on_sale	weekly_amount_sold
0	1	3	12.98	1.0	219.60
1	1	2	12.98	1.0	184.70
2	1	1	12.98	1.0	145.75
3	1	0	12.98	0.0	102.45
4	2	3	19.92	0.0	103.22
5	2	2	19.92	0.0	53.73

where

- `store`: id of the store
 - `weeks_to_xmas`: how many weeks before Christmas the measurement was done
 - `avg_week_sales`: how much they sold in average
 - `is_on_sale`: if store puts product on sale
 - `weekly_amount_on_sale`: how much that store has sold in that week
- Causal inference question: “*what is the impact of T on Y ?*”
 - E.g., what is the impact of `is_on_sale` on `weekly_amount_sold`?
 - E.g., when is the best moment to have sales (i.e., `weeks_to_xmas`)? ^{2 / 24}

Definition

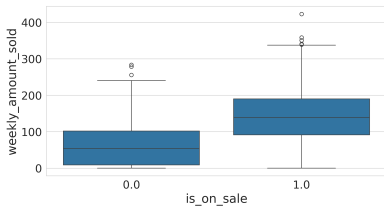
- **Treatment T**
 - $T_i = 1$ if unit i has received the treatment
 - Sometimes T is denoted with D
 - E.g., `is_on_sale` at `weeks_to_xmas`, `is_on_sale`, `received_discount`, `saw_ad`
- **Outcome Y**
 - The variable one wants to influence, Y_i
 - E.g., `weekly_amount_sold`
- **Unit of analysis**
 - The entity for which outcomes are measured



Study	Unit of analysis
Drug experiment	Patient
Marketing experiment	Customer
Sales promotion	Store or product
Education policy	Student

The Fundamental Problem of Causal Inference

- **The Fundamental Problem of Causal Inference**
 - “You can never observe the same unit with and without treatment”
- **Example**
 - Plot outcome
`weekly_amount_sold`
conditioned on treatment
`is_on_sale`
 - Stores with price drops sell more (from 50 to 150)
 - Matches intuition: “people buy more when prices are low due to sales”
- **Is it true?**
 - Maybe only larger businesses afford aggressive price drops?
 - Maybe customers buy close to Christmas anyway?
- Observing both counterfactual situations would reveal the effect of price drops (this is **impossible**)



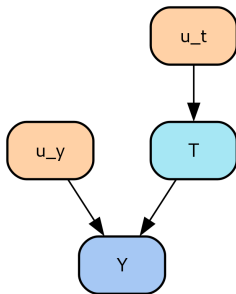
Causal Model

- A **causal model** is a series of assignments $\leftarrow$
 - Causality is non-reversible: you can't rearrange the equation
- **Example**

$$\begin{cases} T \leftarrow f_t(u_t) \\ Y \leftarrow f_y(T, u_y) \end{cases}$$

where:

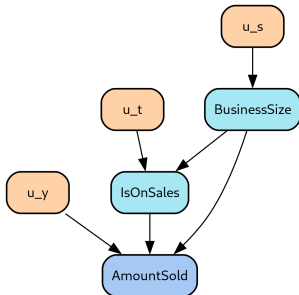
- u_t, u_y are *exogenous* variables (outside the model)
 - * Account for everything not covered by model variables
- Treatment T and u_y cause outcome Y
- T and Y are *endogenous* variables (inside the model)



Causal Model: Sales Example

- Causal model for sales example:

$$\begin{cases} BusinessSize \leftarrow f_s(u_s) \\ IsOnSales \leftarrow f_t(BusinessSize, u_t) \\ AmountSold \leftarrow f_y(IsOnSales, BusinessSize, u_y) \end{cases}$$



- One can make an assumption of linear model for $f_y(\cdot)$ and say

$$AmountSold = \alpha + \beta_1 IsOnSales + \beta_2 BusinessSize + \varepsilon$$

Applying a Treatment to Sales Example

- Assume the **causal model** is:

$$\begin{cases} T \leftarrow f_t(u_t) \\ Y \leftarrow f_y(T, u_y) \end{cases}$$

- Ask “what would happen to the outcome Y if I were to set the treatment to t_0 ?”

$$\begin{cases} T \leftarrow t_0 \\ Y \leftarrow f_y(T, u_y) \end{cases}$$

- The intervention is represented by the $do()$ operator, e.g., $do(T = t_0)$
- Note that **conditioning** and **intervention** are different:

$$\mathbb{E}[\text{AmountSold} | \text{isOnSales} = 1] \neq \mathbb{E}[\text{AmountSold} | do(\text{IsOnSales} = 1)]$$

- The **first** is what happens to businesses that make a price cut
 - * Some business (e.g., big businesses) are “self-selecting” to cut prices
- The **second** is what happens if you force every business to engage in sales

Individual Treatment Effect

- Express the impact of the treatment T on the outcome Y for an individual unit i

$$\tau_i = Y_i|do(T = t_1) - Y_i|do(T = t_0)$$

- The effect τ_i of going from treatment t_0 to t_1 for unit i is the difference in the outcome of that unit under t_1 compared to t_0
 - E.g.,
 $AmountSold_i|do(IsOnSales = 1) - AmountSold_i|do(IsOnSales = 0)$
- You can only observe one term due to the fundamental problem of causal inference
 - Represent it in theory, but can't recover it from data

Potential Outcomes

- Aka “counterfactuals”
- The $do()$ operator represents “unit i outcome is Y if treatment is set to t ”

$$Y_{t,i} \triangleq Y_i | do(T_i = t)$$

- The first is “Rubin’s notation”, the second is “Pearl’s notation”
- For binary treatment use Y_{0i} (without treatment) vs Y_{1i} (with treatment)
 - One is “factual” (i.e., observable)
 - The other is “counterfactual” (i.e., theoretical, not observable)
- There is a slight difference
 - $Y_t = Y | do(T = t)$ is “what a business would sell if it cut prices”
 - $Y | T = t$ is “what a business that cut the prices sold”

Consistency of the Treatment

- There is a single version of the treatment
 - I.e., $T = T_i$ means only one thing
- E.g.,
 - A drug can come in multiple dosages: only one is used
 - Giving a discount coupon is treated as the value of the coupon is the same, it's given only once, in a way that is always equivalent
 - In “help from a financial planner”, is “help” a one-time consultation or regular advice?

No Interference

- Aka “Stable Unit of Treatment Value (SUTVA)”
- The effect of one unit is not influenced by the treatment of other units
 - E.g., there are no spillovers or network effects
- E.g., vaccinating one person makes other close people less likely to catch the illness even if they don't get the treatment

Average Treatment Effects

- Assume you can see both potential outcomes Y_{1i} and Y_{0i} at the same time (which is impossible)
- **Average Treatment Effect (ATE)**
 - The impact that the treatment T would have on the average:

$$ATE = \mathbb{E}[Y_{1i} - Y_{0i}] = \mathbb{E}[Y|do(T = 1)] - \mathbb{E}[Y|do(T = 0)]$$

- E.g., average impact of price cuts on amount sold:

$$ATE = \mathbb{E}[AmountSold_{1i} - AmountSold_{0i}]$$

Conditional Average Treatment Effects (CATE)

- The impact of the treatment on the units in a group defined by $X = x$

$$CATE = \mathbb{E}[(Y_{1i} - Y_{0i})|X = x]$$

- E.g., the impact of having sales during the week of Christmas:

$$CATE = \mathbb{E}[AmountSold_{1i} - AmountSold_{0i} | weeksToXmas = 0]$$

- Like ATE but on a subgroup with a certain characteristics

Average Treatment effect on the Treated (ATT)

- The impact of the treatment on the units that got the treatment:

$$ATT = \mathbb{E}[(Y_{1i} - Y_{0i}) | T = 1]$$

- E.g., how much sales increase for the business engaged in price cuts:

$$ATT = \mathbb{E}[(AmountSold_{1i} - AmountSold_{0i}) | IsOnSales = 1]$$

- We can define these quantities ATE , $CATE$, ATT , but not **measure them** directly
 - Because we can't observe both potential outcomes Y_{1i} and Y_{0i} at the same time

Bias

- **Bias** is what makes association different from causation
- The claim *“cutting prices increases the amount sold by a business”* can be questioned with:
 - *“Those businesses that cut prices could probably have sold more anyway, even with no price cuts”*
- More formally the argument refers to potential outcomes
 - We are trying to estimate $\mathbb{E}[Y_{it}] \triangleq \mathbb{E}[Y_i | do(T = t)]$ using $\mathbb{E}[Y_i | T = t]$ but this is a “biased estimator” (i.e., an incorrect one)

The Bias Equation

- **Causal inference** is interested in:

$$\mathbb{E}[Y_1 - Y_0] = \mathbb{E}[Y|do(T = 1)] - \mathbb{E}[Y|do(T = 0)]$$

- **What is measured** is:

$$\mathbb{E}[Y|T = 1] - \mathbb{E}[Y|T = 0]$$

- Let's express one in terms of the other
 - Replacing the observed outcomes (each unit has two potential outcomes but only one can be observed and the other is 0)

$$Y|T = 1 = (Y_1|T = 1) + \cancel{(Y_0|T = 1)} = Y_1|T = 1$$

- Adding and subtracting $\mathbb{E}[Y_0|T = 1]$, reordering and merging expectations

$$\mathbb{E}[Y|T = 1] - \mathbb{E}[Y|T = 0] = \mathbb{E}[Y_1 - Y_0|T = 1] + \underbrace{\mathbb{E}[Y_0|T = 1] - \mathbb{E}[Y_0|T = 0]}_{\text{Bias}}$$

- Association = causation + a bias term

The Bias Equation: Remarks

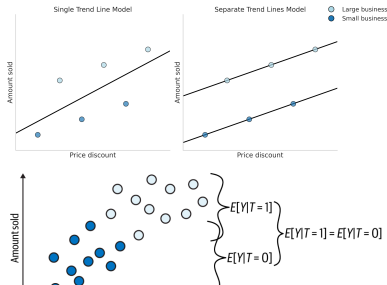
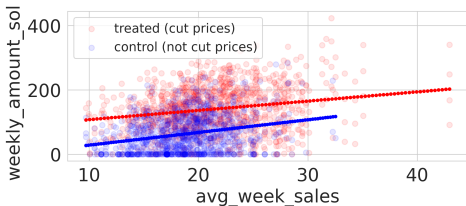
- Association = causation + a bias term

$$\mathbb{E}[Y|T=1] - \mathbb{E}[Y|T=0] = \mathbb{E}[Y_1 - Y_0|T=1] + \underbrace{\mathbb{E}[Y_0|T=1] - \mathbb{E}[Y_0|T=0]}_{\text{Bias}}$$

- Bias arises from differences between treated and control groups (besides the treatment)
 - Unobservable factors change together with treatment
 - * E.g., there are confounders
 - E.g., treated and untreated businesses differ beyond price cuts
 - * E.g., differ in size, location, day of the week on which they have a sale, cities they are located in
- For association to be causation, treated and control units must be **exchangeable**

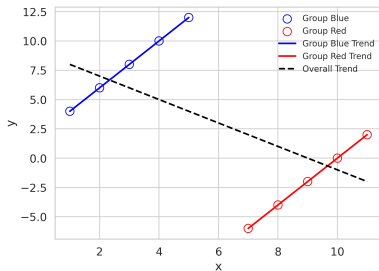
The Bias Equation: Visual Example

- Plot `weekly_amount_sold` by business size (`avg_week_sales`)
 - Color points: treatment (red) / control (blue)
 - Treated units concentrated on the right
- Difference in amount sold between groups has two causes
 - *Treatment effect*: Price cut increases amount sold
 - *Business size*: Bigger businesses sell more, can do more price cuts
- The challenge for causal inference is to untangle the two effects



Simpson's Paradox

- Trend appears in different data groups but reverses when combined
 - Relationship within subgroups can disappear or flip when aggregated
- Paradox occurs due to a confounder influencing the relationship
 - Aggregated data distorts results due to hidden variable



Simpson's Paradox: Admission bias in University

- A university has
 - Two departments: Engineering and English
 - Male overall acceptance rate: $\frac{81}{120} = 67.5\%$
 - Female overall acceptance rate: $\frac{58}{120} = 48.3\%$
- Is a bias against females, due to lower acceptance rate?
- Break down data by Engineering and English

Department	Male			Female		
	Applicants	Accepted	Rate	Applicants	Accepted	Rate
University	120	81	67.5%	120	58	48.3%
Engineering	100	80	80%	20	18	90%
English	20	1	5%	100	40	40%

- At the department level:
 - Women have higher acceptance in Engineering (90% vs 80%)
 - Women have higher acceptance in English (40% vs 5%)
- Paradox reason:
 - Engineering has high acceptance; English has low acceptance
 - More men apply to Engineering; more women apply to English
 - Women apply more to harder department, making their overall

Identifying the Treatment Effect

- We can't observe $\mathbb{E}[Y_{1i} - Y_{0i}]$ since we can only observe one of the potential outcomes Y_{1i} or Y_{0i} and *not both*
- How can we recover causal quantities from observable data?
- The bias equation says that:

$$E[Y|T=1] - E[Y|T=0] = E[(Y_1 - Y_0)|T=1] + \underbrace{E[Y_0|T=1] - E[Y_0|T=0]}_{\text{Bias}}$$

- If $\mathbb{E}[Y_0|T=0] = \mathbb{E}[Y_0|T=1]$, i.e., the treatment and the control group are comparable (exchangeable) regardless of the treatment, then
 - the bias term vanishes $\text{bias} = 0$
 - association is causation
 - the measured difference in means equals $ATE = \mathbb{E}[Y_1 - Y_0]$
 - the measured difference in means equals $ATT = \mathbb{E}[(Y_1 - Y_0)|T=1]$
- **Sales Example**
 - If businesses that do and don't cut prices are similar to each other (i.e., interchangeable), then the difference in amount sold between

Independence Assumption

- The exchangeability property is equivalent to having outcomes being independent of the treatment:

$$(Y_0, Y_1) \perp T$$

i.e., $\mathbb{E}[Y_0|T] = \mathbb{E}[Y_0]$: the treatment gives no information about the potential outcomes

– In other terms $\mathbb{E}[Y_0|T = 1] = \mathbb{E}[Y_0|T = 0]$

- **Sales Example**
 - We can't tell apart businesses that chose to engage in sales from those that didn't

Phases of Causal Inference

- A causal inference problem is often broken down into two steps
 - **Identification**
 - * Express the causal quantity of interest in terms of observable data
 - * Remove the bias, making treated and untreated comparable, so that all the difference can be attributed to the treatment effect

$$\mathbb{E}[Y_0|T = 0] = \mathbb{E}[Y_0|T = 1]$$

- * You need to know how the treatment was distributed or assigned
 - * This is why data alone (observational) cannot answer causal questions
 - **Estimation**
 - * Use data to estimate the causal quantity identified in 1

$$te = \mathbb{E}[Y|T = 1] - \mathbb{E}[Y|T = 0]$$

Randomization removes bias

- Randomization of treatment forces independence to hold and removes bias
- Assume you can randomize the treatment

$$IsOnSales \leftarrow rand(t)$$

$$AmountSold \leftarrow f_y(IsOnSales, u_y)$$

- The dependency to u_t disappears since the assignment mechanism of the treatment is fully known
- Treatment becomes independent from potential outcomes